

LORIS BAZZANI

AI RESEARCH LEADER | MULTIMODAL & GEN AI | ADJUNCT PROF@UNIVR
EX-PRINCIPAL SCIENTIST@AMAZON | PHD

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Current location: Berlin, Germany

MOTIVATION

With a proven track record of scientific publications and transferring science into products, I am excited to take opportunities that push the boundaries of what's possible, break the norms and help growing my technical, leadership and personal skills. Throughout my career, I have been deeply passionate about pushing the boundaries of AI, particularly in the realm of multimodal generative AI, that connect images/videos, language and different types of user interactions (e.g., LMMs, RAGs and stable diffusion). My experience in researching new problems and developing large-scale machine learning and computer vision solutions has equipped me with a unique perspective on leveraging AI to create cutting-edge technology and ensuring that it meets both technical standards and business objectives. One of my key strengths lies in my ability to navigate between scientific breadth and depth, allowing me to effectively reason and contribute to a wide range of scientific work while also diving deep into technical details when necessary. These skills have been instrumental in designing and prototyping complex AI models, articulating their practical value to leadership teams and delivering to production in collaboration with several scientists, engineers, and product managers across different organizations at global level.

EXPERIENCE

University of Verona - Adjunct Professor and Honorary Research Fellow

2025-now

- Part-time position
- Driving research on adaptive and collaborative multimodal intelligence, focusing on the intersection of LMMs and interactive systems.
- Contributing to joint research projects and co-authoring papers with a network of international institutions.
- Teaching the Data Visualization course as part of the Master's degree in Data Science.

Amazon - Principal Applied Scientist

2016-2025

- Research Leadership & Strategy:
 - Defined and led the global science strategy for Large Multimodal Models (LMMs) and Vision-Language Models (VLMs) across Amazon's consumer platforms.
 - Acted as the "Principal Investigator" for multi-year initiatives, securing internal resource allocations (computing and headcount) by presenting research roadmaps to VP- and SVP-level leadership.
 - Pioneered the company's first deep learning models for video understanding in Prime Video, which evolved from an inaugural project into a centralized, cross-functional research program.
- Information Systems & Technical Impact:
 - Architected and deployed production-grade deep learning pipelines for video comprehension, action recognition and automated compliance workflows (e.g., recap of live sports and TV show autotagging for Prime Video).
 - Led the end-to-end development of scalable architectures for multimodal retrieval and generative modeling, serving millions of users on Alexa and Amazon.com and generating multi-million dollar incremental revenue.
 - Translated fundamental research into "Systems-at-Scale", balancing the trade-offs between model complexity (e.g., LMMs and diffusion) and realistic constraints of industrial environments.
- Scientific Maturity & Mentorship:
 - Scaled a centralized science organization from 4 to 15 PhD-level researchers, establishing rigorous peer-review and model-design standards.
 - Mentored 20+ Applied Scientists through direct guidance, promotion cycles and supervised multiple PhD interns from top-tier global universities.

- Advanced the state of the art through publications and patents, serving as a key link between Amazon and the broader academic community.
- Career Progression: Rapidly promoted through three levels of seniority (Applied Scientist → Senior → Principal) based on sustained scientific impact, deployment of research and leadership in Multimodal AI.

Dartmouth College - Postdoc Researcher - Visual Learning Group 2014–2015

- Research on deep learning and computer vision, with production of joint publications and collaboration with industrial entities and international universities, on object recognition and video understanding; and basic teaching activities.

VIBE - Co-founder 2012–2014

- Co-founded a spin-off during my time at the Italian Institute of Technology. VIBE offered technology capable of automatically generating accurate quantitative information about consumer behavior when faced with a physical advertisement. VIBE implemented computer vision and machine learning algorithms that, while strictly respecting privacy, enable the automatic analysis of different consumer categories, their movements, and their reactions.

Italian Institute of Technology - Postdoc Researcher - PAVIS Dept. 2012–2013

- Led research on computer vision and machine learning, participating in industrial projects, with a focus on social interaction analysis, person re-identification and object recognition using dynamic Bayesian nets and Bayesian non-parametric models, and kernel-based methods.

University of British Columbia - Visiting Student - Laboratory for Computational Intelligence 2010

- Explored deep generative neural models, including restricted Boltzmann machines, attentional models, and sequential Monte Carlo methods, laying foundational knowledge that informed later work in deep learning and generative AI.

EDUCATION

University of Verona - Ph.D. in Computer Science Computer 2009–2012

- PhD Thesis: Beyond multi-target tracking: statistical pattern analysis of people and groups
- Supervisors: V. Murino and M. Cristani. Reviewers: A. Del Bimbo and R. T. Collins
- Task leader for some work packages of the European project FP7-217899 SAMURAI

University of Verona - M.S. in Intelligent and Multimedia Systems 2006–2008

Thesis topics: particle filtering and multi-person tracking. Summa cum laude and top of the class

University of Verona - B.S. in Information Technology 2003–2006

Thesis topics: machine learning for medical imaging

RESEARCH & PUBLICATIONS

Multimodal Learning with Interaction/User Feedback

- N. Subedi, L. Bazzani, Z. Al-Halah. Interactive Episodic Memory with User Feedback, [CVPR](#) 2026.
- E. Zorzi, F. Taioli, Y. Wang, M. Cristani, A. Farinelli, A. Castellini, L. Bazzani. Benchmarking Interaction, Beyond Policy: a Reproducible Benchmark for Collaborative Instance Object Navigation, Arxiv 2026.
- Z. Wang, S. Ramasinghe, C. Xu, J. Monteil, L. Bazzani, T. Ajanthan. Learning Visual Hierarchies in Hyperbolic Space for Image Retrieval, [ICCV](#) 2025.
- M. Guillaumin, M. Jaritz, S. Sternig, L. Bazzani. UniCoRN: Unified Commented Retrieval Network with LMMs, Arxiv 2025.
- V. Pippi, M. Guillaumin, S. Casciarelli, R. Cucchiara, M. Jaritz, L. Bazzani. ToFu: Visual Tokens Reduction via Fusion for Multi-modal, Multi-patch, Multi-image Task, Arxiv 2025.
- A. Salvador, E. Gundogdu, L. Bazzani, M. Donoser. Revamping Cross-Modal Recipe Retrieval with Hierarchical Transformers and Self-supervised Learning, [CVPR](#) 2021.
- A. D’Innocente, N. Garg, Y. Zhang, L. Bazzani, M. Donoser. Localized Triplet Loss for Fine-Grained Fashion Image Retrieval, CVPR Workshops 2021.

- Y. Hou, E. Vig, M. Donoser, L. Bazzani. Learning Attribute-driven Disentangled Representations for Interactive Fashion Retrieval, [ICCV](#) 2021.
- Y. Chen, and L. Bazzani. Learning Joint Visual Semantic Matching Embeddings for Language-guided Retrieval, [ECCV](#) 2020.
- Y. Chen, S. Gong, and L. Bazzani. Image Search with Text Feedback by Visiolinguistic Attention Learning, [CVPR](#) 2020.

Multimodal Federated Learning

- A. Chhetri, B. Niroula, P. Shrestha, Y. R. Shrestha, L. A. Anderson, P. K. Gyawali, L. Bazzani, B. Bhattarai. Med-MMFL: A Multimodal Federated Learning Benchmark in Healthcare, Arxiv 2026.

Multimodal Generative Models

- Z. Liu, D. Talon, F. Girella, Z. Ruan, M. Mondo, L. Bazzani, Y. Wang, M. Cristani. Multi-Level Conditioning by Pairing Localized Text and Sketch for Fashion Image Generation, Arxiv 2026 (submitted; under review).
- A. Cao, M. Jaritz, M. Guillaumin, R. de Charette, L. Bazzani. LatteCLIP: Unsupervised CLIP Fine-Tuning via LMM-Synthetic Texts, WACV 2025.
- X. Yang, Y. Zuo, S. Ramasinghe, L. Bazzani, G. Avraham, A. van den Hengel. ViewFusion: Towards Multi-View Consistency via Interpolated Denoising, [CVPR](#) 2024.
- R. Bodur, E. Gundogdu, B. Bhattarai, T-K Kim, M. Donoser, L. Bazzani. iEdit: Localised Text-guided Image Editing with Weak Supervision, CVPR Workshops 2024.
- L. Bazzani, T. Domhan, and F. Hieber. Image Captioning as Neural Machine Translation Task in SOCKEYE, Arxiv 2018.

Video Understanding

- M. Xu, E. Gundogdu, M. Lapin, B. Ghanem, M. Donoser, L. Bazzani. Contrastive Language-Action Pre-training for Temporal Localization, Arxiv 2022.
- L. Bazzani, H. Larochelle, L. Torresani. Recurrent Mixture Density Network for Spatiotemporal Visual Attention, [ICLR](#) 2017.
- V. Murino, S. Gong, C. C. Loy and L. Bazzani, Image and Video Understanding in Big Data, Special Issue in [CVIU](#) 2017.
- L. Bazzani, M. Zanotto, M. Cristani, V. Murino, Joint individual-group modeling for tracking, [PAMI](#) 2015.
- L. Bazzani, V. Murino, and M. Cristani, Decentralized particle filter for joint individual-group tracking, [CVPR](#) 2012.
- M. Denil, L. Bazzani, H. Larochelle, and N. de Freitas, Learning where to attend with deep architectures for image tracking, [Neural Computation](#) 2012.
- L. Bazzani, N. de Freitas, H. Larochelle, V. Murino, J. Ting, Learning attentional policies for object tracking and recognition in video with deep networks, [ICML](#) 2011.
- L. Bazzani, N. de Freitas, J. Ting, Learning attentional mechanisms for simultaneous object tracking and recognition with deep networks, NeurIPS Workshop 2010.
- L. Bazzani, M. Cristani, and V. Murino, Collaborative particles filters for group tracking, ICIP 2010.

Structured Methods for Images

- L. Bazzani, A. Bergamo, D. Anguelov, L. Torresani. Self-taught object localization with deep networks, WACV 2016.
- M. San Biagio, H. Q. Minh, L. Bazzani, V. Murino. Approximate Log-Hilbert-Schmidt distances between covariance operators for image classification, [CVPR](#) 2016.
- H. Q. Minh, L. Bazzani, V. Murino, A unifying framework in vector-valued reproducing kernel Hilbert spaces for manifold regularization and co-regularized multi-view learning, [JMLR](#) 2016.
- H. Q. Minh, M. San Biagio, L. Bazzani, and V. Murino, Kernel Methods on Approximate Infinite-Dimensional Covariance Operators for Image Classification, Arxiv 2016.
- M. San Biagio, L. Bazzani, M. Cristani, V. Murino, Weighted bag of visual words for object recognition, ICIP 2014.

- H. Q. Minh, L. Bazzani, V. Murino, A unifying framework for vector-valued manifold regularization and multi-view learning, [ICML](#) 2013.

Behavioral and Interaction Analysis in Videos

- S. Vascon, and L. Bazzani, Group Detection and Tracking using Sociological Features, Group and Crowd Behavior for Computer Vision 2017.
- L. Bazzani, D. Tosato, M. Cristani, M. Farenzena, G. Paggetti, G. Menegaz, V. Murino, Social interactions by visual focus of attention in a three-dimensional environment, Expert Systems 2013.
- M. Zanotto, L. Bazzani, M. Cristani, and V. Murino, Online bayesian non-parametrics for social group detection, [BMVC](#) 2012.
- L. Bazzani, M. Cristani, G. Paggetti, D. Tosato, G. Menegaz, V. Murino, Analyzing groups: a social signaling perspective, Video Analytics for Business Intelligence 2012.
- M. Cristani, L. Bazzani, G. Paggetti, A. Fossati, A. Del Bue, D. Tosato, G. Menegaz, V. Murino, Social interaction discovery by statistical analysis of F-formations, [BMVC](#) 2011.
- M. Cristani, G. Paggetti, A. Vinciarelli, L. Bazzani, G. Menegaz, V. Murino, Towards computational proxemics: Inferring social relations from interpersonal distances, SocialCom 2011.

Cross-Camera Identification

- L. Bazzani, M. Cristani, and V. Murino. SDALF: Modeling human appearance with symmetry-driven accumulation of local features, Person Re-identification 2014.
- D. Figueira, L. Bazzani, H.Q. Minh, M. Cristani, A. Bernardino, and V. Murino, Semi-supervised multi-feature learning for person re-identification, AVSS 2013.
- P. Salvagnini, L. Bazzani, M. Cristani, and V. Murino, Person re-identification with a PTZ camera: an introductory study, ICIIP 2013.
- L. Bazzani, M. Cristani, and V. Murino. Symmetry-driven accumulation of local features for human characterization and re-identification, [CVIU](#) 2013.
- B. I. Barbosa, M. Cristani, A. Del Bue, L. Bazzani, and V. Murino, Re-identification with RGB-D sensors, ECCV Workshops 2012.
- L. Bazzani, M. Cristani, A. Perina, V. Murino, Multiple-shot person re-identification by chromatic and epitomic analyses, PRL 2012.
- D. S. Cheng, M. Cristani, M. Stoppa, L. Bazzani, V. Murino, Custom pictorial structures for re-identification, [BMVC](#) 2011.
- L. Bazzani, M. Cristani, A. Perina, M. Farenzena, and V. Murino, Multiple-shot person re-identification by HPE signature, ICPR 2010.
- M. Farenzena, L. Bazzani, A. Perina, V. Murino, and M. Cristani Person re-identification by symmetry-driven accumulation of local features, [CVPR](#) 2010.

[Blue color](#) = top-tier computer vision/machine learning conferences or journals.

PATENTS

- US Patent 11,720,942: Interactive retrieval using visual semantic matching, 2023.
- US Patent 11,809,520: Localized visual similarity, 2023.
- US Patent 11,829,445: Attribute-based interactive product recommendations, 2023.
- US Patent 11,416,910: Visual blending of content, 2022.
- US Patent 11,361,212: Machine learning system to score alt-text in image data, 2022.
- US Patent 10,643,074: Automated video ratings, 2020.

CODING & TOOLS

- Organization and planning: Slack, Notion, Asana, ChatGPT

- Coding: Python, PyTorch, HuggingFace, git, bash, LATEX, coding assistants. In past, I worked with MATLAB, Lua/Torch7, JavaScript, Qualtrics, Alexa skill devel tools
- Infrastructure: various AWS services (EC2, Batch, Lambdas, ...) and GPU nodes
- Passionate about and participated to several internal Hackathons (at least 1 per year)

MEMBERSHIPS, AWARDS & GRANTS

- **Member of European Laboratory for Learning and Intelligent Systems (ELLIS)**: selected European scientists that publish regularly in top-tier conferences in AI, 2025.
- **Outstanding Reviewers @ CVPR**, recognition that reflects not only the quality and depth of reviews but also your commitment to fairness, clarity, and timely feedback, 2025.
- **Amazon Accessibility Innovation Award**, internal award given to individuals who created experiences to improve accessibility, 2021.
- **Amazon Robotics Challenge Award**, internal award given to the team who won the robotics challenge, 2018.
- **Finalist at SmartCup Liguria 2013 with VIBE**, 5 startups were selected at the regional level to enter this stage where they were presented to a panel of expert judges, 2013.
- **IBM Best Student Paper Award**, track: Computer Vision at International Conference on Patter Recognition ICPR, 2010.
- **Scholarship** from University of Verona that supported my Ph.D. from Jan. 2009 to Dec. 2011.
- **Scholarship** from EU-Project FP7 SAMURAI, grant FP7-SEC-2007-01 No. 217899, that contributed to support my Ph.D. from Jan. 2009 to Dec. 2011.
- **Travel grant** from University of British Columbia to attend Neural Information Processing Systems, 2010.
- **Travel grant** from International Machine Learning Society to attend the International Conference on Machine Learning, 2011.

AI COMMUNITY EFFORTS

- Co-organizer of the workshop on Human-AI Co-Creation @ ECCV
- Co-organizer of the workshop on Computer Vision for Fashion, Art, and Design @ CVPR/ICCV
- Co-organizer of the workshop on 3D Vision and Modeling Challenges in eCommerce @ ICCV/ECCV
- Co-organizer and program chair of 8 AI conferences and shopping science summits internal at Amazon
- Active reviewer for CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, IEEE TPAMI, IJCV, IEEE Transactions on Multimedia, IEEE Transactions on Image Processing

ITALIAN SCIENTIFIC QUALIFICATIONS

ASN Habitation for Associate Professor in Computer Science (01B1) 2025
 State: obtained the certification given by the Italian Minister of Education reviewed by a national committee.

[Pending] ASN Habitation for Associate Professor in Information Processing Systems (09H1)
 State: applied for this habitation in November 2025 and its approval is pending.

TEACHING & INVITED TALKS

Data Visualization at University of Verona 2025
 Taught the course of Data Visualization which is 48 hours and 6 CFUs).

Machine Learning at Dartmouth College 2015
 Assistant for 10 hours of the undergrad course taught by Prof. L. Torresani.

Introduction to Machine Learning and Computer Vision at the Italian Institute of Technology 2013
 Co-organized this course for PhD students and I taught 6 hours.

Person Re-identification at University of Modena and Reggio Emilia 2011
 Co-organized the course for PhD students at the Short Spring School in Surveillance and taught for 4 hours.

Invited Speaker in Several Venues

Delivered 50+ technical presentations between invited talks by universities, public conferences and internal Amazon events to audiences of up to 1000 people. Selected recent invited talks: “Multimodal Retrieval: From Visual Hierarchies to Interactive and Generative Models” presented at IIT, University of Trento/FBK and Politecnico di Torino in 2025; “Image Generation, Editing and Retrieval with Interactive Elements” presented at Meta AI and the Amazon StatML workshop in 2024.